

Neuromorphic Human-Computer Interaction: A Wikipedia Study

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ABSTRACT

This paper investigates whether human navigation on Wikipedia follows patterns similar to physical movement described by *Fitts' Law*. Using the data from the Wikispeedia dataset, we model navigation as a log-linear relationship between movement time and difficulty of reaching the target article. We explore different metrics for distance and width, which are used to calculate the difficulty, and observe decent correlations. The correlations are strongest when we both semantic similarity and graph structure are combined.

CCS CONCEPTS

• **Computer systems organization;**

KEYWORDS

Neuromorphic Computing, Information Networks, Human Wayfinding, Shannon-Hartley Theorem, Wikipedia

1 INTRODUCTION

Browsing the internet has become an integral part of everyday modern life. The internet is often subdivided into connected information networks that humans navigate using their internal models of the world. Understanding how humans navigate through such information networks gives insight into the connectedness of topics from the viewpoint of users. With this knowledge of semantic relations, the curators of these networks can organise their network nodes in a way that allows for more efficient and intuitive user navigation.

Wikipedia is a well-known example of a virtual information network that can be represented as a directed graph with articles as nodes and links between articles as the edges. It is partially structured by the semantic relatedness of the articles, and therefore acts as an approximation of humans' semantic world models. However, links between articles are added at the discretion of the article's author. While these links are often strongly semantically related, the amount of links and the degree of relatedness between linked articles can vary wildly. This makes it difficult to derive semantic relations from graph structure alone.

Wikispeedia is a game where the goal is for users to navigate from a source to a target article in a small version of Wikipedia [18][17]. Data collected from this game offers insight into the semantic relatedness of articles without direct connections, as well

as the degree of relatedness (the semantic distance) between articles. West et al. [17] demonstrated that semantic distance between concepts can be inferred from this game data.

In the interest of efficient navigation, it is worthwhile to investigate the relationship between the semantic distances of concepts and navigation speed. Understanding how semantic distance affects the time taken to move from a source to a target article offers opportunities to improve graph organisation. For example, rather than only having links added by authors, Wikipedia could provide additional links to concepts that minimise navigation time. This intuition can also be applied to the organisation of other semantic information networks.

We propose that navigation in Wikispeedia can be understood by viewing semantic navigation as analogous to physical movement through space. In motor control tasks, Fitts' law [2] demonstrates a logarithmic relationship between movement time, distance to target, and target width. Wikispeedia navigation strategies tend to follow a two phase structure: initial movement towards a hub article, where large semantic distances are covered quickly, followed by localisation towards the target article where smaller semantic distances are covered over more time. This is similar to the two movement phases observed in Fitts' law tasks: initial fast movement over a large distance towards the target followed by slower more precise movements. This similarity suggests that Wikipedia navigation may follow similar principles to physical movement towards a target and could therefore also be modelled by a log-linear relationship between analogous time, distance, and width metrics.

We investigated this relationship by identifying several possible conceptual mappings for distance, and width in Wikispeedia games to physical distance, and width respectively. We then calculated indices of difficulty using the distance and width mappings and applied linear regression against game duration. We analysed the resulting correlation coefficients to identify if our chosen conceptual mappings exhibit a log-linear relationship.

Can we find conceptual mappings for signal and noise in Shannon's theorem that have a log-linear relationship with the number of clicks?

The paper is structured as follows. In Section 2, we provide background literature that provided us with key insights and motivations. Section 3 the materials we used. In Section 4, we outline our methodology. In Section 5, we present results with brief descriptions. Finally, Section 6 discusses our results and provides insights into our findings. In Section 7 we outline possible future directions and constraints. Section 8 contains conclusions. Our main findings

are that the results suggest that a log-linear relationship exists, but indicate our noise component might not add value.

2 BACKGROUND

In this section we provide an overview of background literature in current academic field. Through this, we aim to embed our project in this field, and present the foundations of our research. We start by outlining theories on human movement in physical and virtual space, and explain our intuitions for extending this to information space, after this we describe conceptualisations of movement and distances in virtual space, specifically focussing on Wikipedia., next we conceptualise distance in targeted search, finally we briefly describe the position of the current paper in the described niche.

2.1 Navigating Physical/Virtual space

2.1.1 Signal-to-noise ratio. Shannon’s Theorem describes the information capacity C (in bits/s) of a communications channel of bandwidth B as

$$C = B \log_2(SN + 1)$$

Fitts proposed an index of difficulty (ID) for a target acquisition task where signal power (S) and noise power (N) are replaced by movement amplitude (A) and target width (W), respectively. This was later reformulated to the Shannon notation, as:

$$ID = \log_2\left(\frac{A}{W} + 1\right)$$

According to MacKenzie “In Fitts’ analogy, movement amplitudes are like signals and target tolerances or widths are like noise.[9]”.

2.1.2 Two phase movement. In a 1996 paper Graham and MacKenzie[4] describe a finding by Woodworth[19] that human pointing movement consists of two phases: “an initial planned impulse which covers most of the distance, followed by a second phase of deceleration to the target under current control”. They find evidence of similar behaviour in virtual pointing behaviour. Virtual pointing is conceptualized as mouse-movement towards a goal, which follows Fitts’ law.

2.2 Navigating Information space

In this paper we investigate navigation in information space, specifically targeted navigation from some starting-page to a goal-page. Specifically, we are interested in investigating whether movement in this information space is similar to movement in physical space. One thing that should be noted is that the navigation in such information networks is restricted as it has to adhere to the graph structure, whereas the movement investigated by MacKenzie[9] is not. A motivation for our hypothesis that such an analogy is sensible is that the two problems seem similar. This is further supported by a similar two-phase movement as described by West et al. [18].

We conceptualize Shannon’s Theorem in information space as follows: movement amplitudes could be computed over some distance over the two objects in the information space, the noise of an object would be the target tolerance akin to Fitts’ law, which in networks would be the ease of access of a node (e.g. PageRank).

2.3 How humans navigate Wikipedia

To use this conceptualisation though, we need to define more concretely how we determine distances for targeted navigation on Wikipedia graphs. An obvious intuition would be to use the shortest path between nodes as the distance, however West et al. note that human behaviour differs from this[17] in targeted Wikipedia navigation. They describe the aforementioned two-phase movement, where users initially move to a “hub”-page, which have a large number of outgoing edges to different topics, after this their movement is mostly guided by semantic features. West et al. later show that human search is very efficient, but uses vastly different navigation than shortest path, with semantic similarity explaining human navigation better[17].

In targeted search we assume the subject is moving towards their goal as fast as they can. In physical space this mean they move the shortest path (on average), however as mentioned above shortest-path graph traversal is not observed in information space. Instead, navigation is guided by semantic similarity. However, humans still move according to a *perceived* optimal way. Since navigation is guided by semantical similarity[18] a network with semantic distances, rather than graph distances might work better.

2.4 Semantic Distances

There are several ways to define semantic distances. Popular baselines exist, such as WordNet or Word2Vec. These are zero-shot models which are known to provide semantic distances between words that are in line with human judgements. There are, however many different ways to find these distances.

2.4.1 Semantic distances from Wikipedia. Several works investigate the possibility of extracting information on the semantic distance between concepts from Wikipedia[3, 6, 8]. They show that useful information can be extracted from the Wikipedia graph structure. The aforementioned papers mostly focus on information that can directly be extracted from the structure of data within Wikipedia. However, there are other aspects to consider as well, such as the way this Wikipedia data-structure is actually utilized by its users. West et al. [18] investigate the possibility of extracting semantic distances from a dataset of Wikispeedia games. This dataset consists of network traversal paths, where each path represents a person playing the Wikipedia game. In this game, a user is tasked to navigate from source-page A to goal-page B in the least amount of steps, thus these traces show the behaviour of targeted navigation on the Wikipedia network. The most relevant finding in this paper is that these traces can be analysed to determine semantic distances between concepts that are sensible when presented to human evaluators, improving over an LSA baseline. This approach was extended in 2013 by Singer et al.[14] to include distances for items which don’t directly occur in the same path, and used a larger dataset. A 2016 paper [10] furthermore finds that even traces from unconstrained navigation can be used to infer semantic similarity between topics.

2.4.2 Semantic distances from transformers. In a 2016 paper, Nogueira and Cho apply these concepts to create a model which produces random walks between nodes on the Wikipedia graph[11]. They reference the work by West et al. stating that they “see that the

goal-driven web navigation is a generalization and re-framing of the Wikispeedia". Their approach places this concept of semantic navigation at its core, representing semantic similarity as a small distance between bag-of-words and attention mechanism embeddings.

In 2022 Zaheer et al. extend the approach outlined by Nogueira and Cho by using transformer models for embedding rather than a bag-of-words approach. Their method vastly improves baselines, at times approaching or improving over state-of-the-art methods. The paper by Zaheer et al. shows strong evidence that semantic similarity extracted by transformers facilitate a good navigation policy "Using random features in place of [transformers] caused the performance to drop to near chance, showing that the models are utilizing the semantics at each node and are not relying on a general search strategy of some kind".

2.5 This paper

Inspired by the Shannon formulation of Fitts' law and the evidence for two-phase, semantically-guided movement in targeted Wikipedia navigation, this paper investigates an analogous relationship for the task difficulty and the number of clicks required to complete a search. We adopt the Fitts' Law analogy, but for the distance component we take into account the semantic distance between the source and goal pages. As human navigation does not follow optimal-graph navigation. The target tolerance is conceptualised as the page-rank. Rather than extracting distance from our data. We do this because otherwise there is a risk of data contamination in our correlation analysis. Zaheer et al. use RoBERTa, we use a small BERT model.

An analysis of user behaviour on Wikipedia is conducted by Piccardi et al. in 2023, and again by Piccardi and West in 2025 [12, 13]. They find that Wikipedia usage does not always follow the same structure as targeted search, meaning results do not generalize to all Wikipedia usage. Specifically natural Wikipedia usage is often very short, only one (or very few) clicks. This means that the results in this paper only apply to targeted navigation on Wikipedia.

3 MATERIALS

In this section we provide a brief overview of the materials used in this project.

3.1 Dataset

The data used in this project originates from the Wikispeedia project [18]. The dataset and more information can be found at <https://dlab.epfl.ch/wikispeedia/play/>.

The following items are relevant for us in this paper:

- articles: a list of all articles
- links: a list of all articles with a link between them
- paths (un)finished: a collection of paths traversing the graph

The graph in this case is defined over the articles as nodes and links as edges.

3.2 Transformer

To create our embeddings we use a BERT based sentence-encoder. Specifically we use the following model on huggingface sentence-transformers/all-MiniLM-L6-v2. It maps sentences and paragraphs to a 384 dimensional dense vector space.

4 METHODS

In order to try and find useful difficulty metrics for Wikispeedia, we compared various potential metrics to see how well they predict the 'real' difficulty, measured by how long the user ended up taking. To do this, we tried the same formula as Fitts' law, $ID = \log_2(\frac{D}{W} + 1)$, by investigating metrics that are conceptually similar to distance (how far we are removed from our target page) and width (how easy it is to reach the target page).

Furthermore, we used binning to group tasks with similar difficulty. The reason for this is that without binning, we have a lot of tasks that are similar in difficulty but have very few samples, causing a lot of noise. We primarily used proportional bins (where we vary bin width and attempt to balance the number of data points per bin) in this paper, but the appendix also contains results without binning (where each datapoint represents a single task with at least three datapoints) and results with fixed-width bins.

After binning, we use linear regression to see if there is a correlation between the index of difficulty and the number of clicks taken. For this, we used SciPy [16].

4.1 Distance metrics

In this paper, we tried the following three distance metrics: *semantic distance*, *semantic graph distance*, and *graph distance*.

The first distance metric, *semantic distance*, is calculated by the LLM transformer model [?] and indicates the similarity of the titles of two Wikipedia articles.

In order to calculate the second distance metric, *semantic graph distance*, we first construct a graph of all Wikipedia pages in the dataset. The articles represent the vertices in the graph, and the links between them are the edges. We then use the LLM's semantic distance as the length of each edge. The semantic graph distance between two articles is the length of the shortest path in this weighted graph.

The last distance metric, *graph distance*, represents the length of the shortest path between two pages, but where all edges simply have a weight of 1, representing the minimal number of clicks.

4.2 Width metrics

The width-like metric we used is *PageRank*. To explain the intuition behind PageRank, assume a user who is surfing Wikipedia. This user starts on some random page and randomly clicks a link. At some point this user gets bored and starts at some other random page and continues clicking links randomly. The PageRank of page a is defined as the chance this user reaches it [1]. Therefore, the higher the PageRank, the easier it is to reach the article, similar to how a wider button is easier to reach, as PageRank grows with more incoming links, and gives more weight to incoming links from pages that are themselves easy to reach.

We also tried using the indegree of a page in the Wikipedia graph, but found that it gave strictly worse results. We will not discuss indegree as a metric further.

4.3 Binning strategies

The dataset we use is quite noisy and contains many entries with slightly different distances and widths. By binning the data on the *ID* we reduce noise and make the relation more smooth. We tried binning the data in two different ways: *fixed width binning* and *proportional binning*.

For fixed width binning, all bins have an equal ‘width’. Since all points are not distributed equally over the *ID*, bins contain big differences in the number of points they contain.

For proportional binning, bins contain the same number of points. Since all points are not distributed equally over *ID*, the ranges of the bins also differ significantly.

5 RESULTS

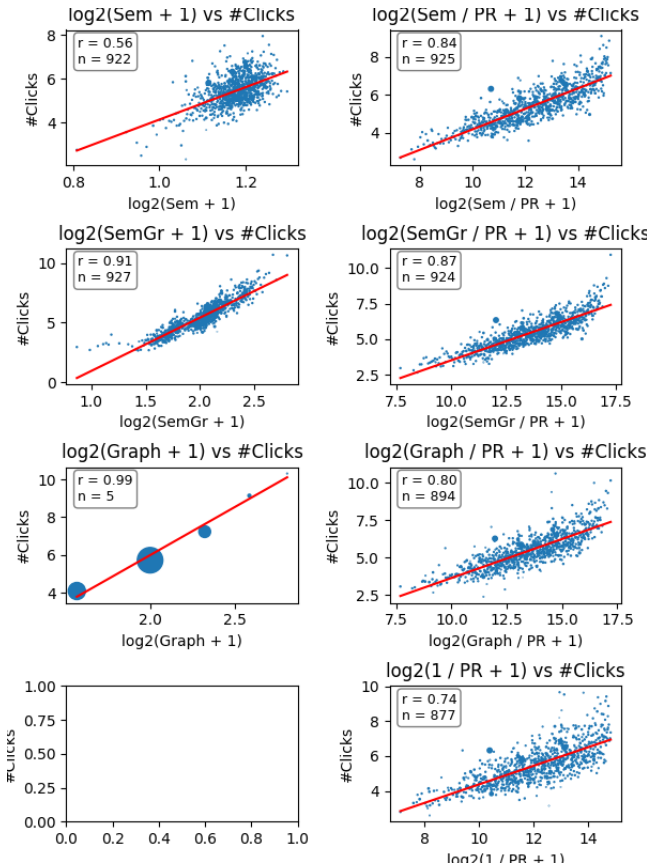


Figure 1: Number of used clicks plotted against difficulty, using various difficulty measures and proportional binning. Left: without PageRank. Right: with PageRank. Top: ‘direct’ semantic distance. Top center: ‘graph’ semantic distance. Bottom center: graph distance. Bottom: without distance metric.

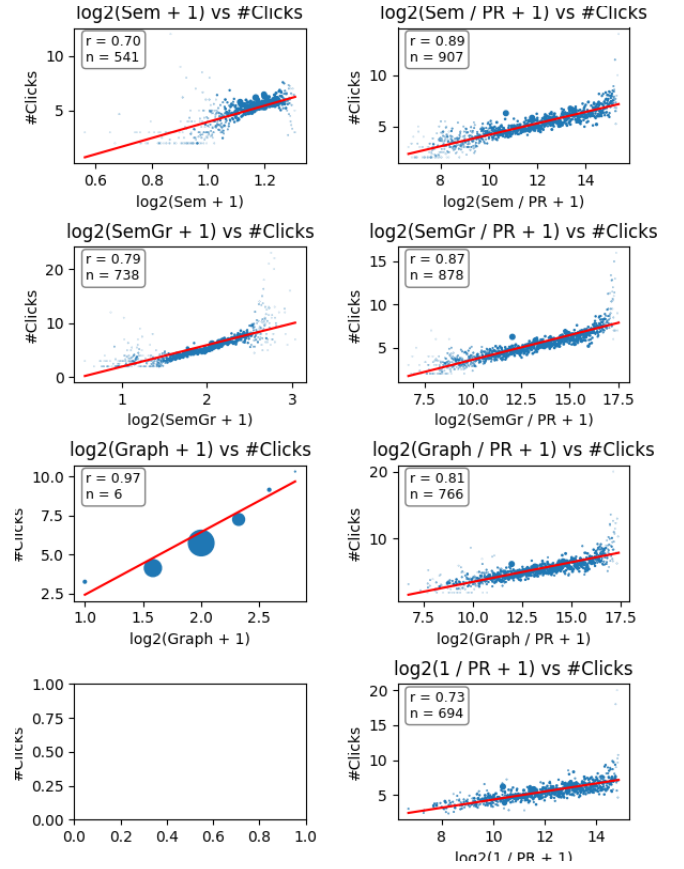


Figure 2: Number of used clicks plotted against difficulty, using various difficulty measures and fixed-width binning. Left: without PageRank. Right: with PageRank. Top: ‘direct’ semantic distance. Top center: ‘graph’ semantic distance. Bottom center: graph distance. Bottom: without distance metric.

We present our results in Figure 1 and Figure 2. These figures differ in the binning strategy used: Figure 1 uses proportional binning and Figure 2 uses fixed size binning. We will go over the various results we have, grouping them by the used distance metric.

Semantic distance. We found that the semantic distance by itself is a relatively poor indicator of task difficulty, with $r = 0.56$ and $r = 0.70$. However, by including the PageRank of the target article, results improved significantly, going up to $r = 0.84$ and even $r = 0.89$. We believe the reason for this is that the semantic distance by itself does not take into account the graph structure, which seems to be an important factor in total task difficulty. Adding PageRank ensures both of these are factored in.

Semantic graph distance. The results here are good, giving $r = 0.91$ for proportional binning and $r = 0.79$ for fixed-size binning without PageRank. This further reaffirms our earlier belief that graph structure is important: this distance metric includes the graph structure by definition. Adding PageRank gives us $r = 0.87$ for both binning methods, which is a relatively small difference, giving a

small improvement for one binning type but a small regression for the other. Overall it seems that adding PageRank here thus does not contribute much.

Graph distance. These results are very limited when we do not include PageRank, as they collapse into 5 or 6 bins. This is because there are only very few possible values here. As such the r -values of $r = 0.99$ and $r = 0.97$ look nice, but are kind of ‘inflated’ by the low number of datapoints. However, it is interesting to see that the relationship does seem to hold according to the limited information we have.

When we add PageRank, which gives us a reasonable number of bins again, we find $r = 0.80$ and $r = 0.81$. This is lower than the other two distance metrics, which seems to indicate that the semantic distance adds useful information about the difficulty of a task.

No distance. Finally, using only PageRank as a difficulty metric gives mediocre results, with $r = 0.74$ and $r = 0.73$. Given that this metric only looks at the target article and not the source article, we think it would be unreasonable to expect better; in fact it is worse than all other metrics we tried, other than semantic distance which ignores the graph structure completely.

Of course, there are no results for not using a distance metric and also ignoring PageRank, as we then do not have a difficulty metric at all. This explains the empty graphs in our figures.

6 DISCUSSION

Metric comparison. Overall, our results seem to indicate that multiple factors are important for the difficulty of a Wikipedia task. The best results are achieved by metrics that take into account a combination of semantic distances and graph structure, while metrics that completely ignore one in favour of the other give worse results.

There is no one best difficulty metric in the results we found, but there are multiple that look promising and seem to give a reasonable indication of the number of clicks a user will require.

Tail end. It also looks like the ‘tail end’ (high difficulty) tasks have a slightly stronger-than-linear upwards trend. However, the data is too sparse and noisy to draw a clear conclusion about this: it can mean that there is no ‘true’ log-linear relationship, but it can also mean that these high-difficulty tasks ‘overwhelm’ the user, and may be similar to discontinuities seen in Fitts’ law [5, 15].

Binning methods. The results for the two binning methods are roughly similar, but differ quite a bit in details. For example, the combination of semantic graph distance and PageRank is the best metric when evaluated using proportional binning, but relatively poor when evaluated using fixed-width binning. Looking at the scatterplots, we hypothesize that this may be related to outliers in these metrics. The graph distance, for example, only has a small number of datapoints with 5 or 6 clicks, and this likely extends to the semantic graph distance having outliers as well. Proportional binning groups these together, thus reducing their impact. On the other hand, semantic distance and PageRank are less likely to have such outliers, which likely explains the good results of this combination when using fixed-width binning.

7 FUTURE WORK

Other metrics. The metrics we discussed, other than semantic graph distance, did not attempt to ‘integrate’ semantic distances and graph structure in one coherent whole. Furthermore, we did not have any width-analogous metrics that takes semantic distances into account, but only PageRank which looks at the ‘raw’ graph structure. Finally, none of our metrics look at the page structure (e.g. how far down a link is), which has been shown to be of importance for navigating Wikipedia [7].

Dataset concerns. Our dataset ‘only’ contains around fifty thousand runs through the game. Furthermore, most source-target pairs only occur a few times in our data, necessitating our approach with binning. As such, it might be appropriate to construct a dataset more geared to this task, e.g. by collecting many runs over a relatively small number of source-target pairs.

Semantic distances. We used only a single embedding model (MiniLM), and only embedded the titles of each Wikipedia article. Potentially, more useful semantic distances can be obtained by using a bigger and newer model, and/or including more data such as the article bodies.

8 CONCLUSIONS

In our project, we attempted to find a difficulty metric for Wikipedia games. We found reasonable success, as long as the metric takes into account a combination of graph structure and semantic distances as given by our LLM embeddings. Metrics that look at these factors in isolation performed worse as a predictor of difficulty.

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A ADDITIONAL INFORMATION

Here we give all results of the experiment. Figures 3 to 5 plot the *ID* against the number of clicks of each game, using proportional binning, fixed size binning and no binning respectively. Figures 6 to 8 plot the *ID* against the duration of each game using proportional binning, fixed size binning and no binning respectively. For each figure, we use the following partitioning. Left: without PageRank. Right: with PageRank. Top: 'direct' semantic distance. Top center: 'graph' semantic distance. Bottom center: graph distance. Bottom: without distance metric.

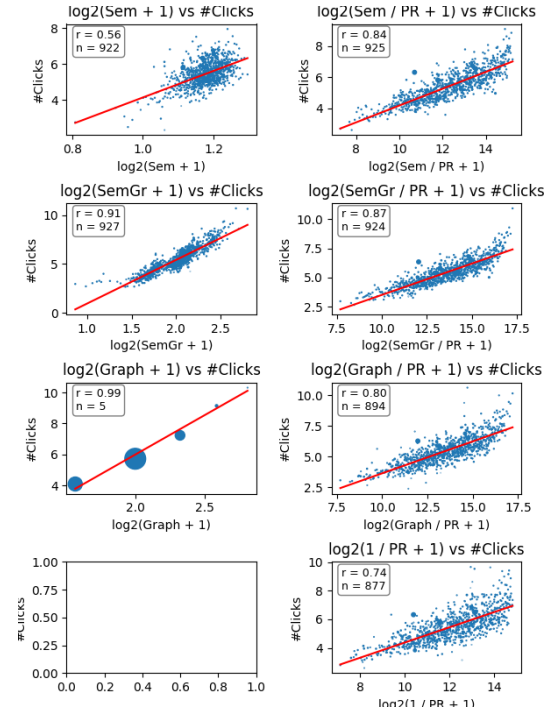


Figure 3: Number of used clicks plotted against difficulty, using various difficulty measures and proportional binning.

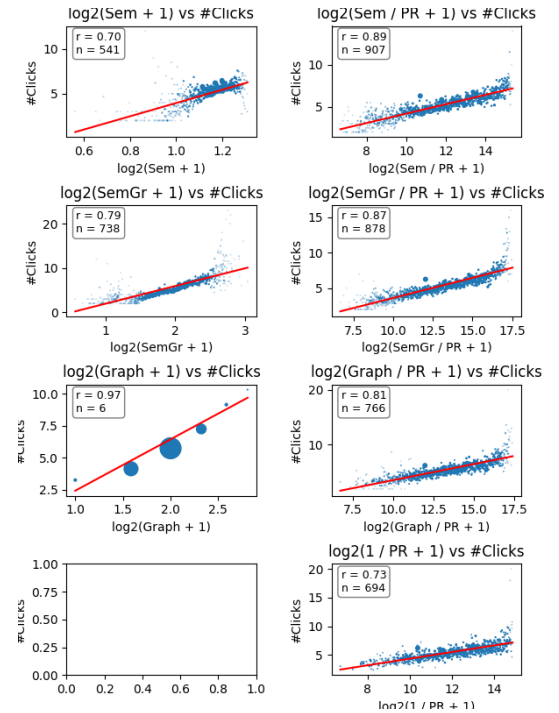


Figure 4: Number of used clicks plotted against difficulty, using various difficulty measures and fixed-width binning.

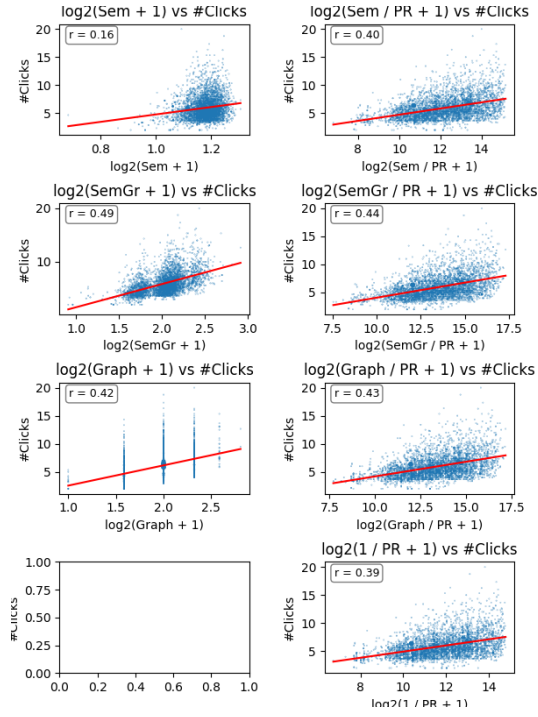


Figure 5: Number of used clicks plotted against difficulty, using various difficulty measures without any binning.

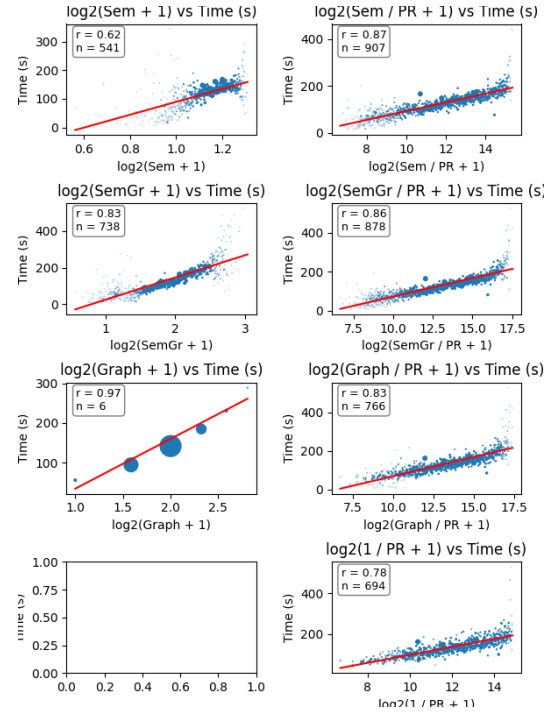


Figure 7: Duration of the game plotted against difficulty, using various difficulty measures and fixed-width binning.

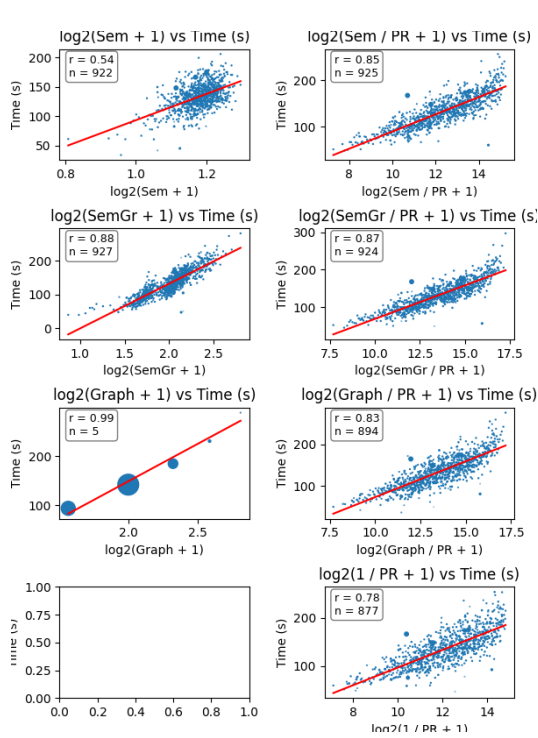


Figure 6: Duration of the game plotted against difficulty, using various difficulty measures and proportional binning.

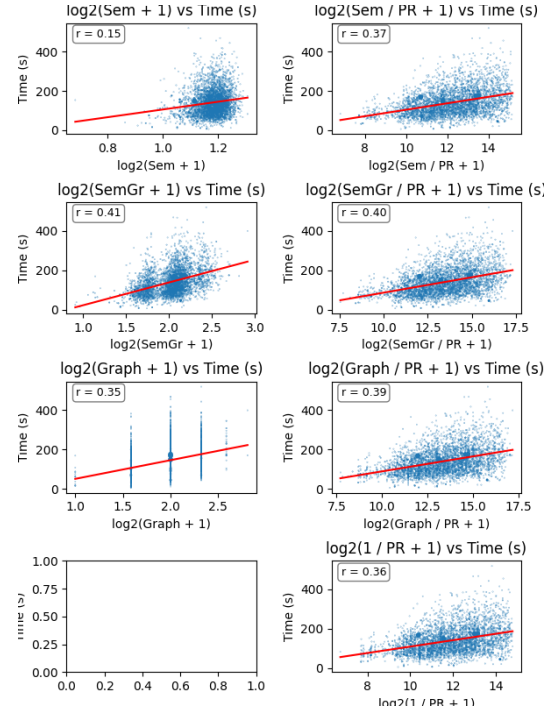


Figure 8: Duration of the game plotted against difficulty, using various difficulty measures without any binning.